

NKOSINATHI NGOZO

Innovative and detail-oriented Full Stack Developer with a strong foundation in software development and computer science. Proficient in designing and deploying scalable web and mobile applications using modern frameworks and technologies such as Node.js, Kotlin, React, and Firebase. Passionate about solving complex problems and delivering user-centric solutions. Seeking to contribute my technical expertise and creative problem-solving skills to a forward-thinking team, driving impactful projects and advancing organizational goals.

CONTACT

Gauteng, South Africa, 0008

(+27) 678-529-440

ngozonathi@gmail.com

www.linkedin.com/in/nkosinathi-ngoza/
ngoza/

<https://nkosinathi-ngoza.github.io/portfolio/>

TECHNICAL SKILLS

- NodeJS
- Kotlin
- C#
- Java
- Python
- React
- Angular
- MongoDB
- Firebase
- Microsoft SQL Server
- Git

SOFT SKILLS

- Problem-solving
- Leadership
- Team Collaboration
- Adaptability
- Time Management
- Organization
- Attention to Detail

EDUCATION

BACHELOR OF COMPUTER AND INFORMATION SCIENCE
APPLICATION DEVELOPMENT - IIE VARSITY COLLEGE

2024

EXPERIENCE

HACKATHON WINNER - IIE VARSITY COLLEGE

2023

WATERFALL SOIT HACKATHON

Overview: Developed a facial recognition-based attendance management system to streamline attendance tracking in educational institutions. The solution efficiently recorded student arrivals and provided real-time notifications to parents, enhancing communication. This innovative approach improved accuracy in attendance records and garnered recognition from industry leaders, showcasing its potential impact on the education sector.

Responsibilities and Roles

- **Team Leadership:** In my role as the lead for a school management system project, I guided the team through the full development process. This included thorough requirements gathering (research) to ensure we met the needs of both students and staff, while ensuring the project stayed on track and met deadlines.
- **Front-End Design:** Created an intuitive front-end interface using Figma, prioritizing a seamless user experience. The designs were focused on making the application user-friendly for both students and staff, ensuring ease of navigation and interaction.
- **Database Management and Facial Recognition:** Utilized MongoDB for efficient student record management and integrated a facial recognition system for automated attendance marking. This system provided real-time email notifications to parents, keeping them updated on their child's arrival at school.
- **AI Integration and Back-End Development:** Integrated OpenAI for generating notes and mini-assignments based on subject topics, and developed an AI-powered quiz creation and grading system. The entire application was built in Python, combining strong back-end functionality with advanced AI capabilities.

- **Collaboration and Presentation Skills:** This project enhanced my skills in collaboration, time management, and communication. I led team discussions, delegated tasks, and ensured deadlines were met. Presenting our solution to judges, including AWS representatives at a hackathon, further strengthened my communication skills, and our project was recognized as one of the winning solutions.

PERSONAL PROJECT – F.G. LATTO

2024

Overview: Developed a comprehensive real estate application designed to simplify property management and transactions for buyers, sellers, and agents. The solution enabled users to explore listings with advanced filtering options, view properties through interactive virtual tours. With integrated communication tools and secure transaction handling, the platform streamlined the buying and selling process while enhancing user experience.

Responsibilities and Roles

- **Client Needs Assessment:** Conducted research and engaged with stakeholders to identify and analyze client needs and pain points, ensuring that application features were tailored to enhance user experience and meet business objectives.
- **Feature Design and Implementation:** Designed and implemented key application features, including user registration, property listings, and evaluation requests, focusing on creating an intuitive user interface that promotes user retention and satisfaction.
- **User Experience (UX) Development:** Developed an engaging and visually appealing user interface that adhered to the F.G. Latto brand identity. Leveraged UX best practices to create a streamlined user journey, from property discovery to transaction completion, optimizing engagement and client retention.
- **Backend Integration:** Designed and implemented a robust API to handle all backend functionality, including user authentication, property data management, secure transaction processing, and communication features. Integrated the backend with a scalable database to ensure efficient data retrieval and storage. Focused on optimizing performance, ensuring data security, and enabling seamless interaction between the frontend and backend systems.

PERSONAL PROJECT – PENNYWISE

2024

Overview: Designed and developed a budgeting and investment mobile application as a personal project to simplify financial management and empower users to make informed decisions. The application featured robust tools for tracking income and expenses, setting goals, and simulating stock investments using real-time data. It also provided an engaging user interface with customizable themes, enhancing the user experience and usability.

Responsibilities and Roles

- **Problem Identification and Solution Design:** The project addressed the challenge of fragmented financial tools by integrating everyday budgeting and investment simulation into a single platform. The app was conceptualized to provide users with tools to track expenses, manage budgets, and simulate stock market trading with a virtual balance, promoting both financial literacy and proactive money management.
- **Development and Technology Implementation:** The app was developed using Kotlin for Android, leveraging Firebase for secure user authentication, real-time data storage, and cloud-based operations. A robust Node.js backend API was implemented to manage database interactions and ensure efficient communication between the app and the server. Key features included a stock trading simulator, customizable themes, and an interactive dashboard for financial insights.
- **User-Centric Features:** PennyWise incorporates a stock investment simulation, where users can trade with a virtual balance of \$10,000, fostering an educational approach to financial growth. The app includes customizable themes (Light, Dark, and Automatic), enhancing accessibility and user preference. The dashboard provides clear insights into spending patterns and budget performance, making financial planning intuitive and actionable.
- **Outcome and Impact:** The completed prototype of PennyWise demonstrated its potential to simplify financial management while educating users on investment strategies. User testing highlighted its intuitive design, engaging interface, and robust functionality, establishing the app as a practical tool for managing personal finances and exploring investment opportunities.